

Mustafa Ozan Karsavuran

Computer Systems Engineer 2, Lawrence Berkeley National Laboratory, Berkeley, California, 94720 USA

✉ ozan.karsavuran@gmail.com | ✉ MOKarsavuran@lbl.gov | 📠 mustafa-ozan-karsavuran-421a7419

SUMMARY

High-performance computing researcher specializing in parallel computing, sparse numerical algorithms, GPU acceleration, and performance optimization. Experienced in developing and optimizing high-performance computational kernels and distributed-memory applications using C/C++, MPI, and OpenMP, with demonstrated speedups on large-scale workloads and architectures. Proven expertise in sparse and irregular computation, communication optimization, and performance characterization across CPUs, GPUs, and large-scale HPC systems. Seeking to apply this expertise to next-generation AI systems and accelerator architectures.

EDUCATION

Bilkent University, Department of Computer Engineering, Turkey

- **Ph.D.** September 2014 – August 2020
- **M.S.** September 2012 – September 2014
- **B.S.** September 2007 – June 2012

PROFESSIONAL EXPERIENCE

Computer Systems Engineer 2

Lawrence Berkeley National Laboratory

November 2025 – *present*

Berkeley, California, USA

- ★Developed a high-performance sparse direct solver for general symmetric linear systems using $A = LDL^T$ factorization and multiple pivoting strategies.
- ★Built a benchmarking framework to compare the solver against MUMPS, MA57, PaStiX, PARDISO, Oblio, and SSIDS, achieving competitive performance with substantially lower memory requirements.
- ★Investigated GPU acceleration and multithreaded BLAS integration.

Postdoctoral Scholar

Lawrence Berkeley National Laboratory

November 2022 – November 2025

Berkeley, California, USA

- ★Developed high-performance sparse direct factorization algorithms for symmetric systems, including positive-definite and indefinite cases, focusing on numerical robustness, fill reduction, and computational efficiency.
- ★Designed and implemented a serial right-looking supernodal sparse Cholesky factorization with reduced memory and data-movement overhead.
- ★Accelerated sparse Cholesky factorization on GPUs, achieving up to $4\times$ speedup; results published at [SC24-W](#).

Postdoctoral Researcher

Bilkent University

September 2020 – October 2022

Ankara, Turkey

- ★Designed and implemented distributed-memory parallel algorithms for sparse matrix/tensor workloads using MPI and communication-aware partitioning.
- ★Developed load-balancing and communication-minimization techniques for distributed stochastic gradient descent, achieving up to achieving up to a $15\times$ speedup.
- ★Scaled sparse tensor decomposition to 4,096 processors and developed hypergraph partitioning methods for large-scale parallel computation.
- ★Benchmarked production HPC applications on six Tier-0 systems using up to 10,000 CPU cores, including architecture-specific builds and performance analysis for NEMO/XIOS.

Teaching and Research Assistant

Bilkent University

September 2012 – June 2020

Ankara, Turkey

- ★Developed and optimized parallel sparse matrix kernels using MPI, OpenMP, AVX, and Intel Xeon Phi, achieving up to 30% runtime reduction.
- ★Designed locality-aware sparse matrix algorithms for many-core processors, achieving up to 20% runtime reduction.

TECHNICAL SKILLS

Systems & Languages: C/C++, Python, Fortran, CUDA, AI-assisted coding

Parallel & Distributed Computing: MPI, OpenMP, Communication Optimization, Load Balancing

GPU & Performance Optimization: Memory Hierarchy Optimization, Data Movement Reduction, MAGMA

Numerical & AI Infrastructure: Sparse Matrix/Tensor Algorithms and Factorizations (LDL^T , Cholesky), Distributed SGD, BLAS/LAPACK

Performance: Benchmarking, Memory Footprint Optimization

PUBLICATIONS

- **M. O. Karsavuran**, E. G. Ng, B. W. Peyton, “Factorization of Sparse Symmetric Indefinite Matrices with Efficient Pivoting”, *technical report in preparation*.
- **M. O. Karsavuran**, E. G. Ng, B. W. Peyton, “GPU Accelerated Sparse Cholesky Factorization” in SC24-W: Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis, Atlanta, GA, USA, 2024, pp. 703-707. doi: [10.1109/SCW63240.2024.00098](https://doi.org/10.1109/SCW63240.2024.00098)
- **M. O. Karsavuran**, E. G. Ng, B. W. Peyton, “A comparison of two effective methods for reordering columns within supernodes”, *arXiv preprint*, 2024. doi: [10.48550/arXiv.2501.08395](https://doi.org/10.48550/arXiv.2501.08395).
- **M. O. Karsavuran**, E. G. Ng, B. W. Peyton, J. Peyton, “Some new techniques to use in serial sparse Cholesky factorization algorithms”, *arXiv preprint*, 2024. doi: [10.48550/arXiv.2409.13090](https://doi.org/10.48550/arXiv.2409.13090).
- K. Büyükkaya, **M. O. Karsavuran** and C. Aykanat, “Stochastic Gradient Descent for Matrix Completion: Hybrid Parallelization on Shared- and Distributed-Memory Systems” in Knowledge-Based Systems, vol. 283, pp. 111176, Jan. 2024. doi: [10.1016/j.knosys.2023.111176](https://doi.org/10.1016/j.knosys.2023.111176)
- N. Abubaker, O. Çağlayan, **M. O. Karsavuran** and C. Aykanat, “Minimizing Staleness and Communication Overhead in Distributed SGD for Collaborative Filtering” in IEEE Transactions on Computers, vol. 72, no. 10, pp. 2925-2937, Oct. 2023. doi: [10.1109/TC.2023.3275107](https://doi.org/10.1109/TC.2023.3275107)
- N. Abubaker, **M. O. Karsavuran** and C. Aykanat, “Scaling Stratified Stochastic Gradient Descent for Distributed Matrix Completion” in IEEE Transactions on Knowledge and Data Engineering, vol. 35, no. 10, pp. 10603-10615, Oct. 2023. doi: [10.1109/TKDE.2023.3253791](https://doi.org/10.1109/TKDE.2023.3253791)
- M. F. Çelikutğ, **M. O. Karsavuran**, S. Acer and C. Aykanat, “Simultaneous Computational and Data Load Balancing in Distributed-Memory Setting” in SIAM Journal on Scientific Computing, vol. 44, no. 6, pp. C399-C424, Nov. 2022. doi: [10.1137/22M1485772](https://doi.org/10.1137/22M1485772)
- N. Abubaker, **M. O. Karsavuran** and C. Aykanat, “Scalable Unsupervised ML: Latency Hiding in Distributed Sparse Tensor Decomposition” in IEEE Transactions on Parallel and Distributed Systems, vol. 33, no. 11, pp. 3028-3040, Nov. 2022. doi: [10.1109/TPDS.2021.3128827](https://doi.org/10.1109/TPDS.2021.3128827)
- **M. O. Karsavuran**, S. Acer and C. Aykanat, “Partitioning Models for General Medium-Grain Parallel Sparse Tensor Decomposition” in IEEE Transactions on Parallel and Distributed Systems, vol. 32, no. 1, pp. 147-159, Jan. 2021. doi: [10.1109/TPDS.2020.3012624](https://doi.org/10.1109/TPDS.2020.3012624)
- **M. O. Karsavuran**, S. Acer and C. Aykanat, “Reduce Operations: Send Volume Balancing While Minimizing Latency” in IEEE Transactions on Parallel and Distributed Systems, vol. 31, no. 6, pp. 1461-1473, June 2020. doi: [10.1109/TPDS.2020.2964536](https://doi.org/10.1109/TPDS.2020.2964536)
- **M. O. Karsavuran**, K. Akbudak and C. Aykanat, “Locality-Aware Parallel Sparse Matrix-Vector and Matrix-Transpose-Vector Multiplication on Many-Core Processors” in IEEE Transactions on Parallel and Distributed Systems, vol. 27, no. 6, pp. 1713-1726, June 2016. doi: [10.1109/TPDS.2015.2453970](https://doi.org/10.1109/TPDS.2015.2453970)

TALKS & POSTERS

- On Solving General Sparse Symmetric Linear Systems, Saint-Girons V, France. (*Presented by E. Ng*)
- Sparse Cholesky Factorization Utilizing GPUs, SIAM CSE25, TX, USA.
- Heuristics for Robust Factorization of Sparse Symmetric Indefinite Matrices, SIAM LA24, Paris, France.
- Sparse Tensor Partitioning for Scalable Distributed CPD-ALS, SIAM PP22, Virtual.
- Medium-Grain Partitioning for Sparse Tensor Decomposition, SIAM CSE21, Virtual.
- Exploiting Matrix Reuse and Data Locality in Sparse Matrix-Vector and Matrix-Transpose-Vector Multiplication on Many-Core Architectures, SIAM CSC16, NM, USA.